

BOX 2026 - Homework 1

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2.

After selecting a chunk of the genome_hw1.fa file of more than 300 lines (without any header within them) of 70 characters per line, and going on nucleotide BLAST, I got a lot of result, all of them which corresponded to ***Streptococcus pneumoniae***. I then tried to check with a different chunk of the file the same procedure, which got me the same result.

3.

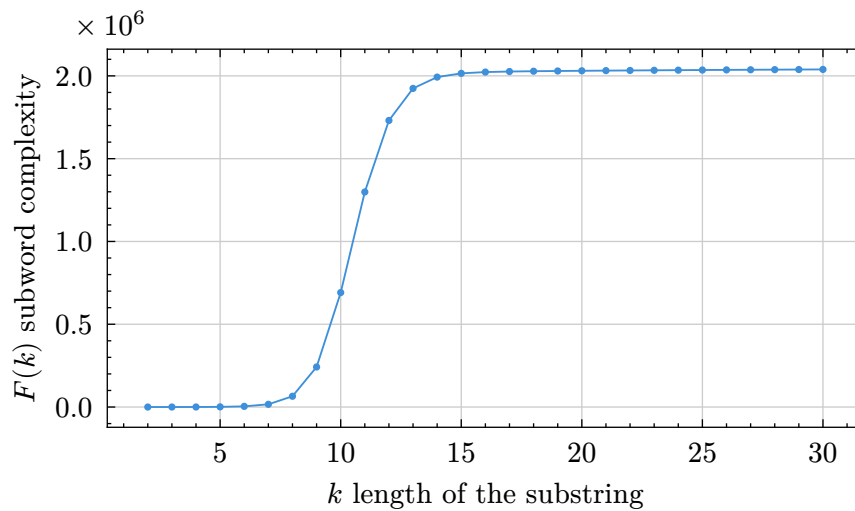


Figure 1: Subword complexity of the base genome

4.

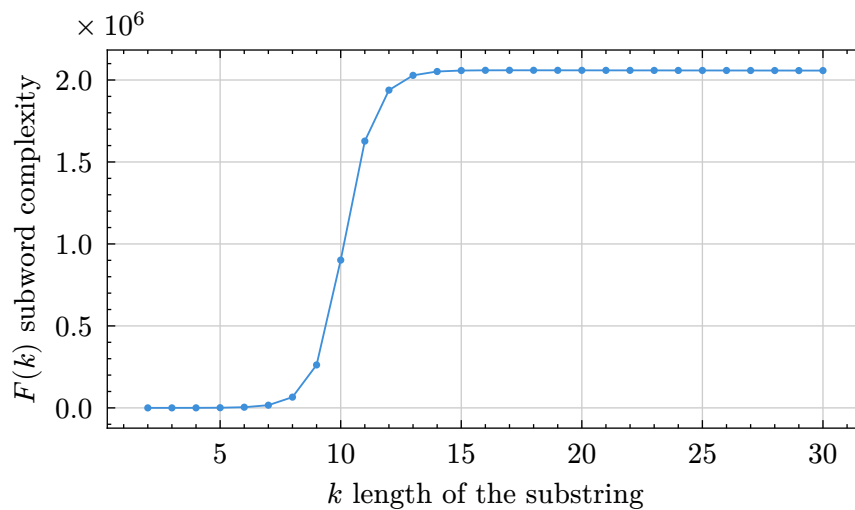


Figure 2: Subword complexity of a similar (in length) random genome

5.

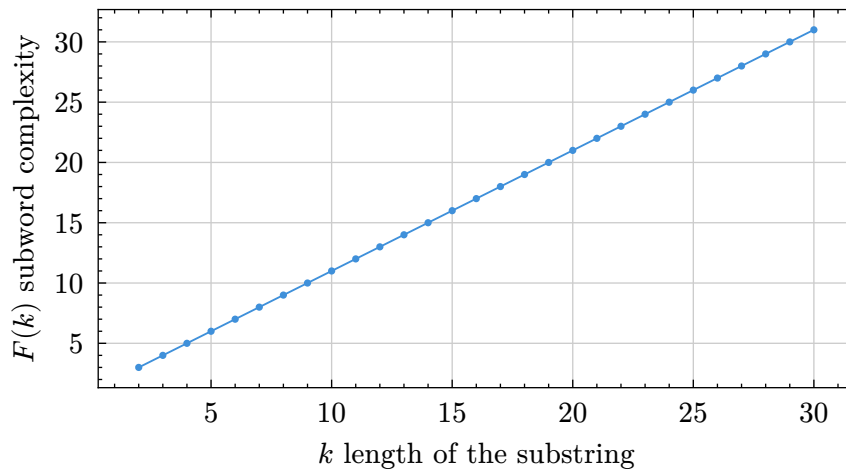


Figure 3: Subword complexity of genome made out of Fibonacci words

6.

Both the genome given with this homework and the random genome we generated based on the previous one share a lot of similarities, which shows that they have a really complex structure :

- At first, the number of different k -mers quickly increase as k increases, and get up to its limit (4^k with $k = 7$ for instance)
- However, the subword complexity then get away of its theoretical limit, because the genomes are still finite, and there are less and less possible position for a k - mer when k increase.
- The only difference between both of them is that the random genomes subword complexity at the end decreases when k increases, while the base genome continues to grow, just at a really slow pace.

At the opposite, the k subword complexity of a genome made out of Fibonacci words increase linearly with k , showing that this genome has an extremely low complexity.